

LLM Scaling Laws

From Kaplan to Chinchilla to Emergent Abilities

Kaplan et al. (2020) | Hoffmann et al. Chinchilla (2022) | Wei et al. (2022)

Introduction

Scaling laws describe how LLM performance improves predictably with model size, dataset size, and compute. Understanding scaling laws is essential for efficient allocation of research resources and predicting capabilities of future models.

1. Kaplan et al. (2020) — OpenAI Scaling Laws

The foundational scaling laws paper showed that loss L follows a power law in N (parameters), D (tokens), and C (compute):

$$L(N) \approx (N_c / N)^{\alpha_N} \text{ where } N_c = 8.8 \times 10^{13}, \alpha_N \approx 0.076$$

$$L(D) \approx (D_c / D)^{\alpha_D} \text{ where } D_c = 5.4 \times 10^{13}, \alpha_D \approx 0.095$$

$$L(C) \approx (C_c / C)^{\alpha_C} \text{ where } \alpha_C \approx 0.050$$

Key finding: model size matters more than data per FLOPs. This led to training very large models (GPT-3: 175B) on relatively small datasets (~300B tokens).

2. Chinchilla (Hoffmann et al., 2022)

DeepMind's Chinchilla paper showed Kaplan's conclusions were wrong due to fixation on compute-optimal training for a fixed budget. Optimal allocation:

$$N_{\text{opt}} = 0.2 \times C^{0.5} \text{ (parameters)}$$

$$D_{\text{opt}} = 5.0 \times C^{0.5} \text{ (training tokens)}$$

Chinchilla (70B, 1.4T tokens) matched Gopher (280B, 300B tokens) on all benchmarks at 4x less compute, demonstrating models should be ~20x smaller but train on ~20x more data.

3. Notable Scaling Results

Model	Params	Tokens	FLOP	Comment
GPT-3	175B	300B	3.1×10^{23}	Undertrained by Chinchilla

Chinchilla	70B	1.4T	5.8×10^{23}	Compute-optimal
Llama-2-7B	7B	2T	1.8×10^{23}	Inference-optimal
Llama-3-8B	8B	15T	—	Over-trained intentionally
GPT-4	~1.8T (MoE)	~13T	$\sim 2 \times 10^{25}$	Estimated
Claude 3 Opus	~2T (est.)	~10T	$\sim 10^{25}$	Anthropic estimate

4. Emergent Abilities

Wei et al. (2022) documented emergent abilities: capabilities that appear suddenly at certain scale thresholds, not predicted by smooth scaling. Examples:

- Few-shot learning: $\sim 10^{22}$ FLOPs (GPT-3 scale)
- Chain-of-thought reasoning: $\sim 10^{23}$ FLOPs
- Instruction following: emerges around 7B+ parameters
- Truthful question answering: improves discontinuously with scale
- Code generation: significant improvement at 12B+ parameters

5. Beyond Next-Token Prediction

Scaling laws for downstream tasks diverge from loss scaling. A model with 2x lower perplexity may be 10x better on some benchmarks (phase transitions). This makes capability prediction from loss alone unreliable for safety-critical applications. Anthropic's model evaluations (evals) are designed to detect emerging capabilities early.